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22 November 1971
SUPERSEDING
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(see 6.5)

MILITARY SPECIFICATION

RADIO SETS AN/SRC-20A AND AN/SRC-21A (INCLUDING RADIO SET AN/URC-9A AND RADIO SET CONTROL C-3866/SRC)

1. SCOPE

1.1 This specification sets forth the requirements applicable to three radio sets which are operable on any of 3500 channels spaced 50 kilohertz (kHz) apart in the frequency range of 225.0 to 399.95 megahertz (MHz) to provide simplex radio communications with aircraft, ships, and shore installations from installations aboard ships and ashore in fixed station installations. The intent is to provide equipment having a high degree of reliability and flexibility in operation. Radio Sets AN/URC-9A, AN/SRC-21A, and AN/SRC-20A operate from a nominal alternating current (a.c.) power source of 115/230 volts, 50 to 60 hertz (Hz), single phase.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Box, Fiberboard

MILITARY

MIL-C-17 - Cables, Radio Frequency, Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead
MIL-P-116 - Preservation, Methods of
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use)
MIL-C-3098 - Crystal Units, Quartz, General Specification for
MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-C-5015 - Connectors, Electrical, AN Type
MIL-Q-9858 - Quality Program Requirements

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MIL-M-10304 - Meters, Electrical Indicating, Panel, Type, Ruggedized
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of
MIL-C-22954 - Control, Radio Set C-1138( )/UR
MIL-C-22959 - Control, Radio Set C-1207( )/UR
MIL-B-23071/16 - Blowers, Miniature, for Cooling Electronic Equipment (10 to 500 CFM), General Specification for
MIL-A-24181 - Amplifier, Radio Frequency AM-1565/URC

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-189 - Racks, Electrical Equipment, 19-Inch and Associated Panels
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-781 - Reliability Tests and Exponential Distribution
MIL-STD-785 - Requirements for Reliability Program

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

DRAWINGS

BUREAU OF SHIPS

RE51F127 - H-172/U Headset-Microphone Assembly.
RE51F128 - H-169/U Handset Assembly.
RE51D121 - Handset, Navy Type 51007A.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification to the extent specified herein.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Ill. 60606.)

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Rules

(Applications for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

3.1 General requirements. The equipment shall meet the requirements of MIL-E-16400, except as otherwise specified herein. The design objectives, as specified in MIL-E-16400 for reliability, simplicity, ease of installation and maintenance consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

3.1.1 GFE. As part of this specification a model will be furnished the contractor as Government-furnished equipment, hereinafter referred to as GFE.

3.1.1.1 GFE. The contractor shall conduct all necessary tests of the GFE to determine electrical conformance to this specification within 90 days after receipt of the GFE. The contractor shall notify the PCO prior to the assembly of first article sample, that the GFE model meets or exceeds this specification. (see 6.2)

3.1.1.2 In the event the GFE, when tested under the conditions of the above paragraph, fails to comply with any of the requirements, it shall be the responsibility of the contractor to include such corrective action as may be necessary for the equipment to be delivered under these specifications to comply with these requirements. The contractor shall notify the PCO in writing of any corrective actions required. (see 6.2)

3.1.2 Temperature. The equipment shall meet the temperature requirements of class 1 of MIL-E-16400, unless otherwise specified herein.

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3.1.3 AN/URC-9A blower assembly. The blower motor assembly shall be in accordance with MIL-B-23071/16.

3.1.4 Electromagnetic compatibility. The equipment shall meet the requirements of MIL-STD-461 in the table in appendix A entitled Requirements Applicable to Reprocurements under MIL-I-16910(C), for equipment class SS-1, except as otherwise stated herein. It shall be the responsibility of the contractor to determine whether any conflict or inconsistency exists between these requirements and the GFE, and upon discovery shall immediately notify the PCO in writing giving full details. (see 6.2)

3.1.5 Parts, materials and processes. Parts, materials and processes shall be in accordance with MIL-E-16400 except as otherwise specified herein. The selection of class, grade, or type shall be such that, when mounted in the equipment, the part or material will perform its intended function under the service conditions of 3.5.1.2. Particular attention shall be paid to the acceptable and unacceptable material requirements of MIL-E-16400.

3.1.5.1 Material and parts. The material and parts shall be of the same quality, grade, type, characteristics, size, and weight as the material used in the GFE, unless otherwise specified herein. The contractor shall use the same parts and material in the first article equipment as in the production units to be delivered.

3.1.5.2 Auxiliary keying relay (k601). An improved version of the auxiliary keying relay shall be provided. The improved version shall have a higher contact current rating. Parts selection shall be in accordance with MIL-E-16400.

3.1.5.3 Squelch relay (K801). An improved version of the squelch relay shall be provided. The improved version shall have better insulation breakdown characteristics between adjacent turns in the coil. Parts selection shall be in accordance with MIL-E-16400.

3.1.6 Parts, subassemblies, and assemblies interchangeability. All parts, subassembly, and assemblies shall be completely mechanically, electrically, and functionally interchangeable with the corresponding part, subassembly, or assembly as used in the GFE unless otherwise specified herein.

3.1.7 Performance. The performance of the equipment shall be within the limits specified herein. In the event of any conflict or inconsistency between the performance or other requirements of this specification and the GFE, the performance or other requirements shall govern. It shall be the responsibility of the contractor to determine whether any conflicts or inconsistency exists and the contractor shall immediately, upon discovery of any such conflict or inconsistency, notify the PCO, in writing, giving full details. (see 6.2)

3.1.8 Design. Unless otherwise specified herein, or specifically approved by the PCO, the electrical design of the equipments to be delivered shall be, in all respects, the same as that of the GFE.

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3.2 Composition of equipment.

3.2.1 Radio Set AN/URC-9A. Radio Set AN/URC-9A shall consist of the following items:

(a) Receiver-Transmitter, Radio RT-581A/URC-9.
(b) Power Supply PP-2702/URC-9.
(c) Case, Receiver-Transmitter Group CY-2959/URC-9.
(d) Cable, Special Purpose, Electrical CX-7260/URC-9 (Receiver-Transmitter Maintenance).
(e) Cable, Special Purpose, Electrical CX-7300/URC-9 (Power supply maintenance).
(f) Cable, Assembly, Special Purpose, Electrical CX-7259/URC-9 (Retransmission Interconnection).
(g) Cable Assembly, Power CX-7258/URC-9.
(h) Kit, Installation, Electronic Equipment MK-620/UR.
(i) A set of special alignment tools in accordance with MIL-E-16400.
(j) A complete set of mating halves of connectors and protective caps in accordance with MIL-E-16400.
(k) Cable Assembly, CX-8521/URC-9A (RT-581/URC-9 Relay Unit Extension).

3.2.1.1 Item 3.2.1(g), (Cable Assembly, Power CX-7258/URC-9), shall not include a locktight-type plug (to mate with the ship's a.c. power) as described on Collins Drawing No. 593-7852-002, but shall include the following type plug: molded rubber plug conforming to Industry No. 5276 with cord grip and finger grip (3 wire, 15 amp, 125 volt a.c. or direct current (d.c.)). The above plug is designed for 115 volt shipboard operation and shall be connected on the one end of the cable. Another connector, molded rubber plug conforming to Industry No. 5676 with cord grip and finger grip (3 wire, 15 amp, 250 volt a.c. or d.c.), shall be provided as part of the cable assembly, but not connected to the cable. The additional connector shall be utilized for 230 volt shipboard operation.

3.2.2 Radio Set AN/SRC-21A. Radio Set AN/SRC-21A shall consist of the following items:

(a) Same items as specified in 3.2.1(a) through (k)
(b) Control, Radio Set C-3866/SRC.
(c) Kit, Installation, Electronic Equipment MK-622/UR (Control, Radio Set C-3866/SRC).
(d) Cable Assembly, Power Electrical CX-6102/U (1 foot, 11-3/4 inches).
(e) Cable Assembly, Special Purpose, Electrical CX-6104/U (2 feet, 5/32 inch).
(f) Rack, Electrical Equipment MT-2300/UR.

3.2.3 Radio Set AN/SRC-20A. Radio Set AN/SRC-20A shall consist of the following items:

(a) Same items as specified in 3.2.2 (a), (b), (c), and (d).
(b) Amplifier, Radio Frequency AM-1565/URC, per MIL-A-24181( ).
(c) Kit, Installation, Electronic Equipment MK-621/UR (Amplifier Radio Frequency AM-1565).

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(d) Cable Assembly, Radio Frequency CG-2232/U.
(e) Cable Assembly, Power CX-6102/U (3 feet, 8-3/4 inches).
(f) Cable Assembly, Special Purpose, Electrical CX-6105/U (1 foot, 6-5/32 inches).
(g) Cable Assembly, Special Purpose, Electrical CX-6105/U (3 feet, 4-5/32 inches).
(h) Rack, Electrical Equipment MT-2299/UR.

3.2.4 Interconnecting cables. Interconnecting cables from Radio Set Control C-3866/SRC to Radio Set AN/URC-9A and Radio Frequency Amplifier AM-1565/URC, as well as the remote control cable between AN/URC-9A and AM-1565/URC (AN/SRC-20A only) shall be in accordance with MIL-C-915, or as otherwise approved by the PCO, and shall be terminated at each end by appropriate AN connectors in accordance with MIL-C-5015.

3.2.5 Electrical equipment racks MT-2300/UR and MT-2299/UR. These racks are designed, because of the weight of the installed equipment, to provide both horizontal and vertical supporting members, together with top and bottom frames to provide for assembly of the two racks listed in 3.2.2 and 3.2.3. Rack, Electrical Equipment MT-2299/UR will normally be provided unless Rack, Electrical Equipment MT-2300 is required by the contract or order. Installation Kit, Electronic MK-512/UR (see 3.2.6) allows conversion of Rack, Electrical Equipment MT-2299/UR to Rack, Electrical Equipment MT-2300/UR. The width of the rack and the horizontal and vertical spacing of the mounting holes on the two front vertical members shall conform to the requirements of MIL-STD-189.

3.2.6 Accessory items. The following items shall be supplied in such quantities as may be specified in the contract or order:

(a) Mounting MT-2554/URC-9 (Provides shock mounting for Case, Receiver-Transmitter Group, CY-2959/URC-9 and contents when mounting on a plane horizontal surface is required).
(b) Mounting MT-2555/SRC (Two required) (Provides shock mounting for Control, Radio Set, C-3866/SRC when mounting on a plane horizontal surface is required).
(c) Mounting MT-2037/URC (Provides shock mounting for Amplifier, Radio Frequency, AM-1565/URC when mounting on a plane horizontal surface is required).
(d) Kit, MK-512/UR (Provides parts to convert Rack, Electrical Equipment MT-2299/UR to Rack, Electrical Equipment MT-2300/UR).
(e) Connectors.
(f) C-3868/SRC Radio Set Control (Permits remote preset channel selection when used with AN/SRC-20A or 21A). The C-3868 shall be in accordance with this specification and BUSHIPS Drawing RE 43D-3074A. It shall not contain radioactive luminous materials.
(g) C-2383/URC Radio Set Control (permits remote preset channel selection and operation of the AN/URC-9A).

3.3 Physical characteristics.

3.3.1 Front panel controls, connectors, and indicators.

3.3.1.1 Receiver-transmitter. The following items shall be located on the front panel of the receiver-transmitter unit:

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(a) Squelch indicator light and disable switch.
(b) Two pilot lamps.
(c) Squelch threshold control (r.f. gain control) and disable squelch switch.
(d) Antenna cable connector (type C, female).
(e) Three position "Mode" switch ("Normal", "Retransmission" and "Tone").
(f) Twenty-one position preset channel selector switch.
(g) Eighteen position manual selector switch (10 mc. steps).
(h) Ten position manual selector switch (1 MHz steps).
(i) Twenty position manual selector switch (0.05 MHz steps).
(j) Twenty position preset channel "memory drum".
(k) Channel number and frequency channel indicators (for preset channel identification).
(l) Two AN Receptacles, MS3102-14S-5S, with caps, in accordance with MIL-C-5015.
(m) One dual self-draining jack for microphone and headset.
(n) Audio gain control.
(o) Test meter.
(p) Meter switch.
(q) Elapsed time meters not required.

3.3.1.1.1 Test meter. The test meter shall conform to MIL-M-10304 and shall provide for checking the proper operation of the equipment as a whole as well as critical circuits. Insertion of the meter into the circuits to be checked, and proper connection of shunts, rectifiers, and multipliers, as needed, shall be accomplished by means of the panel mounted rotary switch. Each switch position shall be marked with the identification of the voltage, current, or other parameter associated therewith.

3.3.1.1.1.1 Antenna circuits. A reflectometer shall be included in the antenna circuits to provide incident and reflected power readings on the test meter which shall adequately indicate the presence of a major fault in the antenna or transmission line.

3.3.1.1.2 Locking devices on controls shall be the same as the GFE.

3.3.1.2 Power supply unit. The following items shall be located on the front panel of the a.c. power supply unit:

(a) Power "on-off" switch.
(b) Power "on" indicator.
(c) Protective fuses, blown fuse indicators are not required.
(d) Spare fuses and lamps.
(e) Dimmer control, power "on" indicator and receiver-transmitter pilot lamps.

3.3.2 Rear panel connectors. The following watertight connectors with caps, in accordance with MIL-C-5015, shall be located on the rear panel of the receiver-transmitter compartment of the Radio Set AN/URC-9A case:

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(a) Remote control.
(b) "Broadband" circuits.
(c) Power input, a.c.
(d) Retransmission.

3.3.3 Controls, internal. The following screwdriver adjustable controls, shall be provided inside the receiver-transmitter:

(a) Carrier squelch range control.
(b) S+N/N squelch control.
(c) Normal mode receive audio gain control.
(d) Receive and transmit compressor control.
(e) Modulator audio gain control.
(f) Retransmit audio output control.
(g) Side tone audio output control.
(h) Broadband side tone adjust.

3.4 General requirements.

3.4.1 Degree of enclosure. The AN/URC-9A shall be watertight in the non-operating condition, and splashproof in the operating condition, in accordance with MIL-STD-108.

3.4.2 Priming and painting. The radio sets shall be primed and painted in accordance with MIL-E-16400. The finish coats shall be light-gray enamel in accordance with type III, class 2 of MIL-E-15090.

3.4.3 Tropicalization. Tropicalization of the complete equipment is not required. Fungus-inert materials in accordance with Requirement 4 of MIL-STD-454 shall be used.

3.4.4 Overload protection. Overload protection shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures of poor adjustments shall have protective devices, or shall be proportioned to withstand such overload without permanent damage to the equipment. The tuning capacitor associated with RF and PA assembly (symbol Z-108) shall be soldered with silver solder, in order to preclude damage due to excessive heat.

3.4.5 Undervoltage protection. The equipment shall not be damaged by voltages below the low limits specified in 3.5.1.2, and shall automatically resume normal operation when the voltage returns within the voltage limits specified in 3.5.1.2.

3.4.6 Broadband. The radio sets shall be provided with the circuits, connector, and relay to make operation with broadband equipment possible.

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3.4.6.1 Broadband operation. When the radio sets are connected to broadband equipment the performance and operation of the radio sets shall be the same as in normal or retransmission mode when receiving or transmitting a plain signal. Reception of a ciphered signal shall cause the plain-cipher relay in the receiver-transmitter unit to be operated by a relay in the broadband equipment. Operation of the plain-cipher relay shall perform the following functions:

(a) Transfer the input of the modular receiver audio amplifier from the plain receiver-transmitter audio circuits to pins of the broadband connector via the associated circuits.
(b) Transfer the receiver-transmitter sidetone output circuit to the broadband sidetone amplifier output.

Transmission and reception shall be in normal or retransmission modes in plain, with the plain-cipher switch on the remote control for broadband equipment in the plain position, and in cipher, with this switch in the cipher position. Operation of broadband plain-cipher switch to the cipher position shall also operate the plain-cipher relay in the radio set by grounding broadband connector lead B.

3.4.6.2 Broadband connector. The broadband connector specified in 3.3.2 shall be Bendix Radio Part No. PTO-7C-12-10-S, or equivalent, and shall provide the following to the broadband equipment:

	Pin	Function
(a)	A	Detector broadband cathode follower output.
(b)	B	Plain broadband relay switching output.
(c)	C	Decoded received audio output (2.19 volts peak-to-peak at 600 ohms).
(d)	D	Spare.
(e)	E	Broadband transmit input (1.5 volts peak-to-peak at 600 ohms).
(f)	F	Common ground.
(g)	G	Microphone output.
(h)	H	Broadband audio sidetone input (50 mw. at 600 ohms).
(i)	J	Spare.
(j)	K	Push-to-talk keying time output (grounded during transmit).

3.4.6.3 Broadband sidetone amplifier. An amplifier shall be provided to raise the transmit audio sidetone out of the broadband equipment from a 50-milliwatt (mw.) level to a minimum of 175 mw. under service conditions. The amplifier shall be capable of operation with output loads from 600 to 120 ohms, corresponding to from one to five 600-ohm headsets. Frequency response of the amplifier shall be within plus or minus 2 db of the 1000-cycle reference from 300 to 3000 cycles under service conditions and a two-headset load. Distortion produced by the amplifier shall not exceed 10 percent under service conditions and a two-headset load (300 ohms).

3.4.7 Shock. Shock requirements shall be for grade A, class II, medium weight, type A of MIL-S-901. Mounting on test machine shall be as approved by the procuring contracting officer (PCO).

3.4.8 Magnetic permeability. The use of magnetic material for external mounting hardware shall conform to the magnetic material requirements of MIL-E-16400. (This material includes hinges, bolts, screws, etc.)

3.5 Performance requirements.

3.5.1 Overall equipment performance. The equipment shall meet all performance requirements specified herein, when operated under standard conditions

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of 3.5.1.1, and shall be capable of consistent and reliable performance under any probable combination of service conditions of 3.5.1.2.

3.5.1.1 Standard conditions. Standard conditions for determination of compliance with performance requirements of this specification shall be as follows :

Supply voltage and frequency	115/230 v.a.c., 50 to 60 Hz, single phase.
Ambient temperature	+20° to +35°C.
Shock, inclination and vibration	None.
Humidity	Prevailing room.
Altitude	Local elevation.

3.5.1.2 Service conditions. Service conditions for determination of compliance with performance requirements of this specification shall be as follows:

Supply voltage and frequency	115/230 v.a.c., ±10 percent, 50 to 60 Hz, ±5 percent single phase.
Ambient temperature	-54° to +65°C.
Shock, inclination and vibration	In accordance with MIL-E-16400.
Humidity	Any value up to 95 percent.
Altitude	Sea level to 10,000 feet.

3.5.2 Radio sets. The radio sets shall be capable of reception and transmission on any of 3500 discrete frequencies spaced 50 kHz apart in the frequency range of 225.0 to 399.9 MHz. It shall be possible to operate on any of these frequencies without changing crystals, reception, and transmission being on the same frequency. Provision shall be made for presetting any 19 frequencies in the 225.0 to 399.9 MHz range by means of the channel selector and the memory drum (see 3.3.1.1) located on the front panel of the receiver-transmitter. The operating frequency of the receiver-transmitter and the number of the preset channel in use shall be indicated by direct reading dials clearly visible at the front panel of the receiver-transmitter unit. Additionally, it shall be possible, by setting the channel selector to a "manual" position, to select any of the available channels by means of the three manual selector knobs. To provide for direct reading frequency indication of any channel selected by the manual selector knobs, the detented positions of these three knobs shall appear on the front panel of the receiver-transmitter as radial lines intersecting at the center of each knob and marked as follows :

(a) 10 MHz selector - 18 positions, marked 22 to 39, inclusive.
(b) 1 MHz selector - 10 positions, marked 0 to 9, inclusive.
(c) 0.05 MHz selector - 20 positions, marked 00 to 95, inclusive.

3.5.2.1 Adjustment and operation. With primary input power applied, it shall be possible to select actual reception and transmission on any of the 19 crystal-controlled preset frequencies by means of the channel selector and to select the same functions at the radio set on any of the available 3500 channels by means of the manual selection knobs when the channel selector is in the "manual" position.

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3.5.2.2 Crystals. The crystals shall be type CR-76/U, type CR-55/U crystals or type CR-18A/U conforming to MIL-C-3098, except as modified to meet the stability requirements of 3.5.4.6.

3.5.2.3 Channel changing cycle. The time elapsing between the instant the channel selector switch is positioned to select a given channel and the instant the equipment is in an operative condition on the selected channel shall be not greater than 8 seconds under standard conditions. This requirement shall hold for shifting the operating frequency of the equipment between any two preset channels. During channel changing operations, the output of the receiver (headset, or handset circuit) shall be shorted or otherwise reduced to zero to indicate, through lack of sidetone, that the equipment is not in readiness for transmission.

3.5.2.4 Transmit-receive interval. When shifting from transmission to reception on the same channel, in "normal" and "retransmission" modes the elapsed time between operation of the control switch and the time the equipment becomes fully operative shall be 325 milliseconds (ms.) or less for local control.

3.5.2.4.1 Receive-transmit interval. When shifting from reception to transmission on the same channel, the time lapse between operation of the control switch and radiation at full power output shall be 60 ms. or less for the AN/SRC-21A and 110 ms. or less for the AN/SRC-20A.

3.5.2.5 Accuracy of automatic tuning. When the same frequency is selected several times in succession, the transmitter r.f. power output and the receiver audio power (with an input signal of 100 microvolts or greater, modulated 30 percent at 1000 Hz) shall not vary by more than 1.0 db from the power output obtained initially.

3.5.2.6 Retransmission. When two equipments are interconnected to allow automatic relaying (retransmission), the performance shall be as follows:

3.5.2.6.1 Single channel communication. When "normal" mode operation, on both receiver-transmitter is selected, performance of each shall be as required by this specification for one receiver-transmitter in a non-retransmission installation.

3.5.2.6.2 Retransmission operation. When "retransmission" mode is selected on both radio sets, the application to the antenna circuit of either radio set of a signal on the receiver resonant frequency and of a level sufficient to open the receiver squelch circuit fully shall cause the other radio set to generate and emit an r.f. carrier wave of the rated power on the selected frequency, and removal of the actuating signal shall cause cessation of the generated carrier. When the actuating signal is 3 db or more above the minimum level at which the foregoing is possible, the generated carrier shall reach rated level within 150 ms. for the AN/SRC-21A or 175 ms. for the AN/SRC-20A after application of the actuating signal, and shall be reduced to substantially zero amplitude within 65 ms. after removal of the actuating signal.

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3.5.2.6.3 Retransmission modulation. When the equipment is relaying in accordance with 3.5.2.6.2 and amplitude modulation is applied to the actuating signal, the emitted carrier shall be modulated in an essentially similar manner. When a signal 3 db above the level required to actuate the squelch (or 10 microvolts, if the squelch opens on noise) is applied and is modulated 80 percent at 1000 Hz, the emitted carrier shall be modulated at least 80 percent.

3.5.2.6.4 Retransmission monitor. When the equipment is relaying as specified in 3.5.2.6.2, the operator's headset or handset shall be connected so as to monitor all signals relayed in either direction, and the operator's microphone shall be connected so as to actuate and modulate both transmitters when he closes the switch. Under these conditions, the audio power output, distortion, and signal-plus-noise-to-noise ratio at the terminals of the headset circuit and the carrier output and modulation shall be essentially the same as those obtained when "normal" operation is selected.

3.5.3 Receiver.

3.5.3.1 Receiver input impedance. The receiver radio frequency (r.f.) input impedance shall be nominally 50 ohms.

3.5.3.2 Dummy antenna. For test purposes, the r.f. input to the receiver shall be through a 50-ohm (resistive) dummy antenna. This value of 50 ohms shall include the internal impedance of the signal generator. Should this latter impedance exceed 5 ohms, the r.f. input levels to the receiver shall be corrected accordingly.

3.5.3.3 Sensitivity . The sensitivity of the receiver shall be such that over the frequency range of 225.0 to 399.95 MHz, and r.f. input of 6 microvolts or less, modulated 30 percent at 1000 Hz, shall produce an audio output of at least 50 milliwatts (mw.) with a signal-plus-noise-to-noise ratio (modulated to unmodulated) of 10 db under standard conditions (see 3.5.1.1) and with the "normal" mode of operation selected.

3.5.3.4 Automatic gain control (a.g.c) characteristics. With the front panel volume control set to maximum clockwise position and an r.f. input signal of 100 microvolts modulated 30 percent at 1000 Hz applied to the receiver input through the dummy antenna of 3.5.3.2, the internal audio level limit control shall be adjusted to provide an audio output level of 2.0 watts; the front panel volume control shall then be set to reduce the audio output level to 500 mw. Under this condition, when the r.f. signal input, 30 percent modulated at 1000 Hz is varied from 10 to 250,00 microvolts, the audio output power shall not vary by more than ±2 db from the audio output obtained from the 100-microvolt signal. The receiver shall also be protected against blocking under conditions of r.f. input up to 1 volt.

3.5.3.5 Selectivity. The overall selectivity of the receiver shall fall within the following limits:

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Attention below resonant response Decibels (db)	Total bandwidth of receiver kHz
6	80 minimum
60	150 maximum

3.5.3.6 Images and other spurious responses. When receiving on any channel, the receiver shall attenuate images and other spurious responses not less than 60 db, relative to the desired signal, except that all spurious responses within plus or minus 2 MHz of the resonant frequency shall be attenuated not less than 40 db.

3.5.3.6.1 Internal signals. Internal signals generated by oscillator interaction and harmonics falling on the desired frequency or within the pass band shall not cause interference on more than 10 channels within the frequency range. Such interference shall not degrade the operating sensitivity more than 6 db.

3.5.3.7 Oscillator radiation. The limit for test CE06 of MIL-STD-461 shall be 51 db microvolts.

3.5.3.8 Intermediate frequency (i.f.) rejection. The receiver shall present an attenuation of not less than 100 db, relative to the responses of the desired signal on the frequencies of the i.f. amplifiers.

3.5.3.9 Regeneration. The interstage and overall regeneration of the receiver shall be so minimized that no evidence of unnecessary oscillation, instability, or excessive regeneration shall occur at maximum r.f. gain setting.

3.5.3.10 Noise peak limiter. The receiver shall be provided with an instantaneous series noise peak limiter which shall function continuously in normal and retransmission modes of reception, but shall not affect receiver operation when broadband operation is selected. The noise limiting action shall effectively limit or reduce voltage surges or peaks exceeding the receiver output obtained with a 45- to 65-percent modulated signal at maximum sensitivity. Its action shall be relatively instantaneous so as not to obliterate signals by a permanent reduction in receiver gain under influence of continuous r.f. noises such as are caused by commutators and ignition systems. When a pulsed signal having a pulse duration of 1 microsecond, a pulse repetition frequency of approximately 500 pulses per second, and peak amplitudes up to 0.1 volt is applied on the same frequency as a desired signal modulated at 30 percent at approximately 1000 Hz, not more than 50 microvolts input (desired signal) shall be required to obtain a signal-plus-noise-to-noise ratio of 3 db. The noise limiter threshold shall be set by the a.g.c. voltage automatically, and shall adjust itself to accommodate signals of different or varying signal strengths. The noise limiter shall be automatically disabled when "Broadband" operation is selected by the main-cipher relay.

3.5.3.11 Cathode follower output. The output of the audio frequency (a.f.) detector shall be fed directly to a cathode follower whose output is available at a receptacle on the rear panel. The cathode follower shall be designed for

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an output load impedance of 600 ohms. With a carrier displaced ±12 kHz from the receiver resonant frequency and 30 percent modulated with audio frequencies from 300 Hz to 25 kHz, the frequency response shall be as follows:

Frequency (Hz)	Maximum attenuation (db)
300	3
1,000	0 (reference)
10,000	2
15,000	4
20,000	5
25,000	7

Phase distortion shall be held to an absolute minimum consistent with the frequency response.

3.5.3.12 Receiver audio characteristics.

3.5.3.12.1 Receiver output impedance. The receiver a.f. output impedance shall be 600 ohms.

3.5.3.12.2 Dummy load. For test purposes, the audio output shall be terminated in a 600-ohm resistive load, which shall include means for measuring the a.f. power output.

3.5.3.12.3 Power output. The receiver shall be capable of delivering an audio output of at least 2.0 watts when a signal having any level between 100 and 100,000 microvolts, modulated 30 percent at 1000 Hz, is applied and the total harmonic content of the receiver shall not exceed 10 percent at the 2.0-watt level. Since the modulator of the transmitter also functions as the audio amplifier of the receiver, a screwdriver adjustable internal level control shall be incorporated to allow the audio output to be set to the 2.0-watt level.

3.5.3.12.4 Receiver audio compressor. Adjustment of the modulator compressor in the transmit condition (see 3.5.4.4.3.1), so that compression has just started when 80-percent modulation is obtained, shall automatically set the operating point of the receiver audio compressor. The output of the receiver audio amplifier shall be 39 volts ±1 db across a 600-ohm load when the input to the receiver is 100 microvolts, modulated 30 percent at 1000 Hz, and the receiver audio amplifier is adjusted as previously specified. A 6-db increase in input to the receiver audio amplifier or an increase in load impedance to 1200 ohms shall not cause the output to vary more than ±2 db from the reference of 39 volts ±1 db.

3.5.3.12.5 Overall fidelity. The audio response to the receiver audio amplifier shall be:

300 Hz ±5 db
500 Hz ±4 db
1000 Hz 0 db
3500 Hz ±4 db

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3.5.3.12.6 Audio gain control. Operation of the audio gain control on the radio set shall cause a minimum of change in the impedance which the radio set presents to external equipment attached to the headset or handset terminals, and such impedance shall be not less than 50 ohms at any setting of the audio gain control.

3.5.3.13 Carrier level squelch. The r.f. gain control (squelch threshold control) shall have such range as to permit a variation in the sensitivity of the receiver from the maximum to a value of not less than 100 microvolts. This control, together with suitable carrier level squelch circuits provided in the receiver, shall permit adjustment so as to suppress unwanted signals or noise in the receiver output under conditions of no or very low (depending upon the setting of the control) r.f. input signals to the receiver. With the r.f. gain control adjusted so as to obtain at least 50 mw. output with an r.f. input of 6 microvolts, modulated 30 percent at 1000 Hz, the audio output of the receiver shall be effectively suppressed for all r.f. inputs, similarly modulated, not in excess of 3 microvolts on the same operating frequency. Operation of the squelch circuit shall be positive, requiring at its threshold of operation a change of approximately 1 db in r.f. input modulated 30 percent at 1000 Hz for effecting a change in audio output of at least 10 db.

3.5.3.13.1 Carrier level squelch. All AN/URC-9A radio sets shall be delivered with the associated circuitry connected for carrier level squelch operation.

3.5.3.14 Signal-plus-noise-to-noise ratio squelch. In addition to the r.f. carrier squelch circuit, a squelch circuit shall be provided which will be actuated whenever the signal-plus-noise-to-noise ratio exceeds a predetermined level adjustable by means of an internal control. The squelch circuit shall be capable of being adjusted to operate on signals which provide a signal-plus-noise-to-noise ratio of from 3 to 10 db. Operation of this squelch circuit shall be substantially independent of changes in r.f. gain such as are caused by change of frequency, line voltage, and tube aging. The attack time shall not exceed 75 ms., and the release time shall be between 1.0 and 5.0 seconds, when the squelch is set to open with a signal-plus-noise-to-noise ratio of 3 db. For measurement of attack and release time an r.f. test signal modulated 30 percent at 1000 Hz and at a level which provides a 10-db signal-plus-noise-to-noise ratio at the receiver output shall be used. The signal-plus-noise-to-noise ratio squelch shall be selected automatically by the mode switch when in the "normal" mode and "tone" mode positions only. Conventional carrier level squelch shall be selected automatically when "retransmission" operation is selected. Carrier level squelch or signal-plus-noise-to-noise ratio squelch may be made available for both the "normal" and "tone" modes of operation by repositioning one lead on the Modulator subassembly. When the unit is in broadband operation, the receiver audio output is taken from the broadband cathode follower and shall not be squelched. Broadband operation is selected by means of a broadband control unit.

3.5.4 Transmitter.

3.5.4.1 R.f. power output. The transmitter shall be capable of delivering at least 16 watts and no more than 24 watts of r.f. carrier energy to a dummy load (see 3.5.4.3) at any frequency within its range under standard conditions. When the AN/URC-9A is delivered as part of the AN/SRC-20A, under standard conditions, the r.f. power output of the AN/SRC-20A shall be at least 100 watts and no more than 130 watts into a dummy antenna. The dummy antenna shall be as specified in 3.5.4.3.

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3.5.4.2 Transmitter output impedance. The r.f. output impedance shall be nominally 50 ohms.

3.5.4.3 Dummy load. For test purposes, the r.f. output load (dummy antenna) shall consist of 10 feet of transmission line conforming to type RG-213/U of MIL-C-17, terminated in a 50-ohm resistive load which shall include means for measuring the r.f. power output percentage modulation, and distortion.

3.5.4.4 Modulation characteristics. The transmitter shall meet the following performance requirements for modulation:

3.5.4.4.1 Modulator input impedance. The modulator input impedance shall be nominally 82 ohms.

3.5.4.4.2 Dummy microphone. For test purposes, the audio (microphone) input circuit of the transmitter shall be connected to an a.f. generator and dummy microphone having the following characteristics:

Resistance	82 ohms.
Impedance (mainly resistive at audio frequencies)	82 ohms.
Frequency range	150 to 20,000 Hz.
Total harmonic distortion	Less than 5 percent
Open-circuit output potential (adjustable)	0.01 to 3.0 volts root mean square (r.m.s.)
Meter, a.c. (to indicate open-circuit voltages, regardless of load)	0 to 3.0 volts r.m.s.
Meter, d.c.	0 to 100 milliamperes (ma.).

3.5.4.4.3 Radio telephone (A-3).

3.5.4.4.3.1 Modulator capabilities. The modulator of this equipment shall employ a compressor circuit which shall control the gain of the speech amplifier stage in the modulator. With the mode switch in the "normal" mode and "retransmission" mode positions, the modulator shall be capable of symmetrically modulating the r.f. carrier output of the transmitter not less than 80 percent, when the open circuit output voltage at 1000 Hz obtained from the dummy microphone (see 3.5.4.4.2) is adjusted to approximately 0.08 volt and the compressor is inoperative. When the input level and the compression control are adjusted so that the compressor has just started to apply bias to the grid of the speech amplifier, a 12-db increase in input to the modulator shall cause not more than 2-db increase in output. The compression control shall have sufficient range of adjustment to permit operation with no compression at 80 percent modulation, or with full compression at 64 percent modulation. When the input voltage at 1000 Hz obtained from the dummy microphone is adjusted to produce 80-percent modulation with no compression, the input level shall be between 0.04 and 0.12 volt. With the input voltage maintained at a fixed level approximately 3 db less than the above determined level, the transmitter shall have the following fidelity characteristics relative to the output at 1000 Hz.

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Frequency (Hz)	Relative limit (db)
300	+3
1000	0 (reference level)
3500	+3

3.5.4.4.3.2 Modulator "broadband" fidelity. When operating in the "broadband" mode the carrier modulation shall have the following fidelity characteristics relative to the output at 1000:

Frequency (Hz)	Relative limit (db)
300	+0.0, -3.0
1,000	0.0 (reference level)
10,000	+1.0
25,000	+0.0, -6.0

3.5.4.4.3.3 Distortion. With input and compression adjusted as specified in 3.5.4.4.3.1 for the determination of frequency response, the total harmonic content of the modulation envelope shall not exceed 7.5 percent.

3.5.4.4.3.4 Noise modulation. Noise modulation of the transmitter carrier shall be at least 35 db below the maximum modulation obtainable, and the receiver signal-plus-noise-to-noise ratio shall be at least 30 db when a 10,000-microvolt signal modulated 30 percent at 1000 Hz is applied, and the total receiver output (signal plus noise) is adjusted to 500 mw.

3.5.4.4.3.5 Frequency modulation. Under the service conditions and with no modulating voltage applied to the transmitter and no r.f. signal applied to the receiver, the peak frequency deviation arising from modulation of the receiver conversion oscillator or the transmitter carrier shall not exceed 1.0 kHz. Under standard conditions, and with the emitted carrier modulated 80 percent at 1000 Hz, the peak frequency deviation arising from frequency modulation of the transmitter shall not exceed 1.0 kHz.

3.5.4.5 Electromagnetic compatibility. The transmitter shall meet the applicable requirements of 3.1.4 except for the following:

3.5.4.5.1 Spurious radiation. The magnitude of all radiated signals on other than the carrier frequency, exclusive of harmonics and its primary sidebands, shall be more than 60 db below the carrier level when the transmitter operating frequency is in the frequency range of 245 to 380 MHz. When the transmitter is operating on any frequency outside the preceding frequency range, a maximum of one spurious radiated signal shall be permitted with a magnitude less than 60 db below the carrier level. In all cases the level of any such spurious radiation shall be more than 30 db below the carrier level.

3.5.4.5.2 Radiated emission from cabinet. Electromagnetic radiation from the equipment exclusive of intended radiating members, shall meet the limit requirements of test RE02 in accordance with the figure entitled Limit for RE02 Narrowband Emissions and the figure entitled Limit for RE02 and RE05 Broadband Emissions of MIL-STD-461.

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3.5.4.5.3 Harmonic radiation. Radiated signals harmonically related to the carrier frequency except the second harmonic shall be at least 60 db below the carrier level. The second harmonic of the carrier frequency shall be at least 50 db below the carrier level, when the transmitter carrier frequency is in the frequency range of 240.0 to 399.95 MHz, inclusive, and at least 30 db below the carrier level when the transmitter carrier frequency is in the 225.0 to 240.0 MHz frequency range.

3.5.4.6 Frequency stability. The combined effects of temperature and humidity, and mode of operation shall result in transmit and receive frequencies of:

+150°F.	+12 kHz	-40°F.	+15 kHz
+100°F.	±10 kHz	-65°F.	±20 kHz
Ambient	±10 kHz		

Transmission and reception on each channel shall be possible with the resulting transmitter carrier frequency and receiver each within the allowed tolerance of the nominal frequency for such channel. The frequency shall not drop more than 3.5 kHz when the nominal supply voltage is decreased to 103 (or 207) volts, a.c. The frequency shall not raise more than 3.0 kHz when the supply voltage is increased to 127 (or 253) volts, a.c.

3.5.4.7 Sidetone power. Monitoring the carrier modulation shall be effected via the receiver audio output channel regardless of the position of the mode switch. The modulation signals delivered to the headset or handset shall be obtained from the receiver output winding on the modulation transformer. The sidetone power available shall be adjustable from 0 to at least 175 mw. when the transmitter carrier is modulated 80 percent at 1000 Hz. The total harmonic content of the output shall not exceed 10 percent.

3.5.4.8 Audio accessories. The following audio accessories are representative of those normally used during A3 communication:

Item	Bureau of Ships Drawing	FSN
(a) Headset - H-172/U Microphone	RE51F127	F5965-715-4031
(b) Handset - H-169/U	RE51F128	N5965-679-9501
(c) Handset - NT-51007A and CX-1846/U cord assembly	RE51D121	N5965-665-3480 cord assembly supplied separately

These items will be supplied as Government-furnished property in the quantity as specified in the contract or order.

3.5.4.9 Emmission tone (A-2). Selection of the "Tone" position of the mode switch shall disable the microphone line and enable a 1000 cycle tone oscillator. Closing the key line shall cause the radio set to emit a carrier, tone modulated not less than 70 percent. Total harmonic content of the modulation envelope shall not exceed 10 percent. Receiver characteristics for "Tone" operation shall be the same as for "Normal" mode.

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3.5.5 Power supply. The power supply shall operate from an a.c. power source of 115/230 volts, 50 to 60 Hz (nominal), single phase, to provide the following voltages at the required power level:

(a) +325 volts d.c., +155 volts d.c., for electron tube circuits.
(b) 6.3 volts a.c. for electron tube heaters.
(c) 26.5 volts d.c., for frequency selection and relay operation.
(d) 115 volts a.c., for blower motors.

3.6 Control, Radio Set C-3866/SRC.

3.6.1 General description and purpose. This radio set control shall be provided for operation of Radio Set AN/URC-9A, Radio Frequency Amplifier AM-1565/URC, and an antenna multicoupler, by means of Control Indicator C-3868/SRC, in conjunction with control, Radio Set C-1138/UR or C-1207/UR ("Radio-phone" unit) conforming with MIL-C-22954 and MIL-C-22959, respectively. Functional demands made by as many as four "Radiophone" units located throughout a ship shall be transferred by control, Radio Set C-3866/SRC to the transmitting-receiving equipment.

3.6.2 Physical characteristics.

3.6.2.1 Overall size and weight. The following dimensions and weight shall not be exceeded:

Width	Depth	Height	Weight
19 inches	19-1/2 inches	8-1/2 inches	68 pounds

3.6.2.2 Construction and mounting. The style of construction and material shall be as used in Radio Set AN/URC-9A insofar as practical. Drawer type slides shall be provided to support the unit when withdrawn from the case for maintenance. Handles shall be provided to assist in withdrawing and inserting the unit from and into the case. The shipboard radio set control shall provide for mounting in a special 19-inch rack, or for plane surface mounting. Detachable shockmounts shall be provided for plane surface mounting, if required, to allow the unit to meet stated requirements pertaining to shock and vibration.

3.6.2.3 Degree of enclosure. This unit shall be splashproof in accordance with MIL-STD-108.

3.6.2.4 Operating controls, indicators, and connectors.

3.6.2.4.1 The following controls and indicators shall be provided on the front panel of the control unit:

(a) Emergency power switch.
(b) Momentary contact switch (on-off).
(c) Local-remote switch.
(d) Two power indicator lamps.
(e) Telephone dial (11 position).
(f) Fuseholders.

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(g) Spare fuses and indicator lamps.
(h) Nineteen squelch potentiometers (accessible upon opening front panel door).

3.6.2.4.2 The following connectors conforming to MIL-C-5015 shall be mounted on the rear of the control unit:

(a) Radio set control unit (radiophone connector).
(b) Channel selector (indicator unit connector).
(c) A.c. power input connector.
(d) Antenna multicoupler connector.
(e) AN/URC-9A control connector.
(f) AN/URC-9A power connector.
(g) AM-1565/URC power connector.
(h) Receiver audio output connector.

Connectors mating with the connectors listed shall be supplied with the unit, except for connectors terminating interconnecting cables, supplied with the equipment.

3.6.3 Electrical characteristics.

3.6.3.1 Power input control and indication. Power input to control, Radio Set C-3866/SRC, shall be 115/230 volts a.c. +10 percent, 50/60 Hz +5 percent, single phase. All power for Radio Set AN/URC-9A and Radio Frequency Amplifier AM-1565/URC shall be controlled by an emergency power switch and a start-stop switch located on the front panel of the shipboard radio set control and also by as many as four start-stop switches located remotely throughout the ship. All main and branch power circuits shall be adequately fused for protection against overload. A panel mounted indicator lamp shall be illuminated when the emergency power switch is closed. A second panel mounted indicator lamp shall be illuminated when any of the five start-stop switches are closed. The a.c. power line shall not be grounded in any manner.

3.6.3.2 Channel control circuits. It shall be possible to dial and obtain proper operation on any one of 19 channels (preset on Radio Set AN/URC-9A, Radio Frequency Amplifier AM-1565/URC, and an antenna multicoupler from a telephone type dial and associated apparatus specified in 3.6.1).

3.6.3.2.1 Dialing interference. Random interference generated during the channel dialing operation at the local position C-3866 or remote C-3868 shall be exempt from measurement under requirements of MIL-STD-461 provided that the duration of the interference does not exceed 1.2 seconds.

3.6.3.3 Squelch control circuitry. A separate squelch control shall be provided for each of the 19 channels. Selection of the correct squelch control for the channel selected shall be accomplished by means of the telephone stepping switch also utilized for channel control and indication.

3.6.3.4 Audio and transmit-receive control circuitry. Transmit-receive control (push-to-talk), receiver audio, sidetone, audio, and transmit audio.

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passing between the remotely located control stations and the receiving-transmitting equipment shall be channeled through C-3866/SRC Control unit.

3.6.3.5 Power supply. A power supply shall be included which shall furnish a nominal 12 volts d.c. with not more than 0.5 percent ripple voltage. It shall be capable of supplying a continuous load of 660 ma. without sign of overheating. Temporary overloads for 1 minute, at infrequency intervals, of 1.2 amperes (external load) shall not cause damage. The output voltage shall be within the limits of 9.5 to 15 volts from the overload conditions to minimum load and for specified variations of line voltage. Both terminals of the 12-volt d.c. power supply shall be brought out to connector pins to supply microphone voltage or current, push-to-talk control to the radio set, and carrier-on indication at as many as four "Radiophone" units located aboard the ship. Convenient means shall be provided for grounding the positive side of the 12-volt d.c. power supply. The 12-volt d.c. output shall be isolated from the main power supply line.

3.6.3.6 Microphone circuit. Means shall be provided for matching the microphone input circuit of Radio Set AN/URC-9A to the 600-ohm output of "Radiophone" units.

3.6.3.7 Local-remote operation. A local-remote switch shall be incorporated in control, Radio Set C-3866/SRC to disable all associated "Radiophone" units and permit local operation only.

3.7 Reliability.

3.7.1 Quantitative reliability requirements. The AN/SRC-20A, AN/SRC-21A, and AN/URC-9A radio sets shall each have a minimum acceptable mean-time-between-failure (MTBF) of 500 hours.

3.7.2 Reliability program plan. The contractor shall prepare for approval a reliability program plan in accordance with the requirements of this specification and the requirements of MIL-STD-785. The reliability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the reliability program (see 6.2).

3.7.3 Reliability prediction. A design reliability prediction shall be performed for the AN/SRC-20A, AN/SRC-21A, and AN/URC-9A radio sets in accordance with the procedures described in MIL-HDBK-217 based on stress factors and part populations. All mechanical parts shall be included in the reliability prediction. The prediction shall be based on the operational requirements of this specification for a shipboard environment with an ambient temperature of 65°C. The reliability mathematical model and reliability block diagram shall be submitted with the reliability prediction report. The source and applicability of all failure data used for the reliability prediction shall be documented in the reliability prediction report. (See 6.2.)

3.7.4 Reliability testing. Reliability testing shall be performed in accordance with 4.5.

3.7.5 Reliability reports. The contractor shall prepare a report on the procedures, results, and conclusions for the reliability activities of 3.7 and the reliability program plan (see 6.2).

3.8 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

3.9 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirement. The contractor shall provide and maintain a written quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

4.1.2 Quality plan. A quality plan, applicable to the equipments covered by this specification, shall be prepared for Government approval (see 6.2).

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; (b) Government product inspection to measure quality of product to be offered for acceptance; and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to correct promptly deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Test procedures. The contractor shall prepare test procedures for Government approval covering first article and quality conformance inspection (see 6.2).

4.1.5 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

(1) Production inspection (see 4.4.1).
(2) Production control inspection (see 4.4.2).
(3) Environmental inspection (see 4.4.3).
(4) Inspection of preparation for delivery (see 4.4.4).

(c) Reliability testing (see 4.5).

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4.3 First article inspection. Unless otherwise specified (see 6.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine conformance with the requirements of this specification. First article inspection shall include the examination and testing specified in production inspection, production control inspection, environmental inspection, and the additional inspections listed in table I.

Table I. Additional first article inspections.

Inspection	Reference
Power	MIL-E-16400
Shock	MIL-E-16400
Vibration	MIL-E-16400
Inclination	MIL-E-16400
Salt spray test	MIL-E-16400

4.3.1 First article inspection report. A first article inspection report shall be submitted to the Government covering the results of the contractor's first article inspection (see 6.2).

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests and examinations approved in accordance with 4.1.4 and shall include the requirements of table II under standard test conditions (see 3.5.1.1).

Table II. Production inspection.

Inspection		Reference
(a)	Receiver	
(1)	Sensitivity	3.5.3.3
(2)	A.g.c. characteristics	3.5.3.4
(3)	Selectivity	3.5.3.5
(4)	Images and other spurious responses	3.5.3.6
(5)	Regeneration	3.5.3.9
(6)	Noise peak limiter	3.5.3.10
(7)	Cathode follower output	3.5.3.11

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Table II. Production inspection. (Continued)

Inspection	Reference
(8) Audio power output	3.5.3.12.3
(9) Receiver audio compressor	3.5.3.12.4
(10) Overall fidelity	3.5.3.12.5
(11) Carrier level squelch	3.5.3.13
(12) Signal-plus-noise-to-noise ratio squelch	3.5.3.14
(b) Transmitter	
(1) R.f. power output	3.5.4.1.
(2) Modulator capability and fidelity	3.5.4.4.3.1
(3) Modulator broadband fidelity	3.5.4.4.3.2
(4) Distortion	3.5.4.4.3.3
(5) Noise modulation	3.5.4.4.3.4
(6) Frequency stability (under ambient conditions)	3.5.4.6
(7) Sidetone power output	3.5.4.7
(c) Radio Set Control C-3866( )/SRC	
(1) Channel control circuits	3.6.3.2
(2) Squelch control circuitry	3.6.3.3
(3) Audio and transmit-receive control circuitry	3.6.3.4
(4) 12 volt d.c. power supply	3.6.3.5
(5) Microphone circuit	3.6.3.6
(6) Local-remote operation	3.6.3.7
(d) In-process inspections	4.4.1.1
(e) Operating test	4.4.1.2
(f) AN/URC-9 blower assembly	4.4.1.3
(g) Cables	4.4.1.4

4.4.1.1 In-process inspections. The following in-process inspections shall be required:

(a) Workmanship, soldering, assembly, fit and marking.
(b) Materials, parts, and finish.
(c) Treatment for prevention of corrosion and tropicalization.

4.4.1.2 Operating test. The equipment, assembled to include all parts required for delivery of end items, shall be energized and subjected to an operating test to ensure qualitatively the correct functioning of the equipment, including all external operating controls and exercise of all relays. Test meter (see 3.3.1.1.1) functions shall be included.

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4.4.1.3 AN/URC-9A blower assembly. The blower assembly shall be inspected to determine conformance with 3.1.3.

4.4.1.4 Cables. The cables listed in 3.2 shall be subjected to a continuity test and a voltage test. The continuity test shall ensure that wires terminate on the correct connector pins. The voltage to be applied to each cable for the voltage test shall be equal to the highest voltage rating without exceeding the voltage rating of any constituent part of the cable assembly. Leakage current during the voltage test shall not exceed one milliampere.

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.4. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment. Lot formation shall be as defined in MIL-STD-109.

Table III. Production control inspection.

Inspection	Reference	AQL%
Supply line voltage and frequency	4.4.2.3	6.5
Heat test	4.4.2.4	6.5
Enclosure	4.4.2.5	6.5
Channel changing cycle	3.5.2.3	6.5
Transmit-receive internal	3.5.2.4	6.5
Receive-transmit internal	3.5.2.4.1	6.5
Accuracy of automatic tuning	3.5.2.5	6.5
Retransmission	3.5.2.6	6.5
Image response	3.5.3.6	6.5
I.f. rejection	3.5.3.8	6.5
Audio gain control	3.5.3.12.6	6.5
Frequency modulation (standard conditions)	3.5.4.4.3.5	6.5
Harmonic radiation	3.5.4.6	6.5
Emission tone	3.5.4.10	6.5
Parts, Subassemblies, and Assemblies interchangeability	4.4.2.6	6.5
Magnetic Permeability	3.4.8 (see note)	6.5

Note: Magnetic Permeability shall be preformed on one sample unit from the first production control lot and need not be reperformed.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to, and have passed, production control inspection may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all visible damage.

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4.4.2.3 Supply line voltage and frequency test. Supply line voltage (115 and 230 volts) and frequency (115 volts) tests shall be performed in accordance with MIL-E-16400. The following tests shall be performed under service conditions (see 3.5.1.2). Reference measurements shall be made under standard conditions (see 3.5.1.1) in order to establish conformance with the following requirements:

4.4.2.3.1 Receiver.

4.4.2.3.1.1 Audio power output (see 3.5.3.12.3) shall not be reduced by more than 4 db below the level measured under standard voltage and frequency conditions for any specified variation of voltage and frequency.

4.4.2.3.1.2 Sensitivity (see 3.5.3.3) shall not be degraded by more than 3 db by any specified voltage or frequency variation.

4.4.2.3.2 Transmitter.

4.4.2.3.2.1 R.f. output (see 3.5.4.1) shall not be reduced by more than 3 db below the level measured under standard voltage and frequency conditions for the specified variation.

4.4.2.3.2.2 The channel changing cycle (see 3.5.2.3) shall not exceed 15 seconds at the -10 percent voltage level.

4.4.2.3.2.3 The transmit-receive interval (see 3.5.2.4) shall not exceed 75 ms. for the AN/SRC-21A and 140 ms. for the AN/SRC-20A. Radio sets shall be operated in the "normal" mode.

4.4.2.3.3 Radio Set Control C-3866/SRC. The radio set control shall be operated to control Radio Set AN/URC-9A channel selection by means of the front panel telephone dial and transmit-receive operation of the radio sets through appropriate test instrumentation.

4.4.2.4 Heat test. The heat test shall be performed in accordance with MIL-E-16400. The radio sets shall be keyed to transmit continuously throughout the test. Suitable remote instrumentation shall be utilized for operation of the equipment during the test. Output power of the radio set and receiver sensitivity shall be measured at the end of 8 hours and prior to reducing chamber temperature. Measurements shall be made at 19 preset channels approximately equally spaced throughout the operating range. At the end of the test, the radio set shall be visually examined for any degradation that may have occurred.

4.4.2.5 Enclosure test. The enclosure test shall be performed in accordance with MIL-E-16400 and the applicable paragraphs of this specification specifying the degree of enclosure.

4.4.2.6 Parts, subassemblies, and assemblies interchangeability. The major parts and assemblies, listed in subparagraphs below, shall be subjected to examination and test to verify complete interchangeability. The manufactured part or assembly shall be mechanically substituted in a corresponding equipment and the examination or test performed as indicated. The major parts and assemblies shall be substituted one at a time within an equipment.

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4.4.2.6.1 Part, subassembly, or assembly

a. Radio Set AN/URC-9( )
   1. RF and PA assembly
   2. Frequency multiplier oscillator assembly
   3. First IF amplifier assembly
   4. Second IF amplifier assembly
   5. Third IF amplifier assembly
   6. Relay-Filter assembly
   7. Front panel assembly
   8. Audio amplifier and modulator assembly
   9. Filter assembly
   10. Centrifugal fan assembly [RT-581( )]
   11. Frequency selector assembly
   12. Directional coupler
   13. Broadband sidetone amplifier assembly
   14. CY-2959/URC Equipment Cabinet, Electrical
   15. PP-2702/URC-9( ) Power Supply
   16. RT-581( )/URC-9( ) Receiver-Transmitter, radio
   17. CX-7258/U; CX-7259/U; CX-7260/URC-9( ); CX-7300/URC-9( ); and CX-8521/URC-9( ) cable assemblies
   18. MK-620/UR, Kit, installation, electronic equipment.

b. Control, Radio Set, C-3866/SRC
   1. MK-622/UR, Kit, Installation, Electronic equipment

c. Amplifier, Radio Frequency, AM-1565/URC

   1. Cabinet, Electrical equipment
   2. Power amplifier assembly
   3. Servo amplifier assembly
   4. Chasses and front panel assembly
   5. Blower assembly
   6. MK-621/UR Installation Kit

d. Radio Set AN/SRC 20( ) and AN/SRC-21( )
   1. CG-2232/U; CX-6102/U; CX-6105/U(1.5 feet); CX-6105/U(3.5 feet); CX-6105/U
   2. MT-2300/UR Rack, electrical equipment
   3. MK-622/UR Installation kit, Electronic equipment
   4. MT-2299/UR Rack, electrical equipment

e. Field change kits and miscellaneous items.

4.4.2.6.2 Each first article equipment shall be subject to the interchangeability requirements. For first article equipment, the interchangeability requirements shall be verified using the Government Furnished equipment. As a quality conformance inspection, one sample equipment from each lot shall be subjected to the interchangeability requirements and the interchangeability shall be between the approved first article equipment and the end item manufactured equipment. If first article inspection is not required, the interchangeability shall be between corresponding end item manufactured equipment.

4.4.2.6.3 Examination and tests. The contractor shall select tests and examinations from production inspection to verify the interchangeability requirements. Tests and examinations shall be selected on the basis of being meaningful to the particular part, subassembly, or assembly interchanged. Where mechanical criteria are not specified for mechanical items, the contractor shall provide sufficient detail to verify the mechanical aspects of interchangeability.

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Table IV. Environmental inspection.

Inspection	Reference
Temperature	4.4.3.4
Humidity	4.4.3.5
Radio interference	4.4.3.6

4.4.3.1 Sampling for environmental inspection. For the first environmental inspection, one equipment from the first month's production or from the first 50, whichever is less, shall be selected. For subsequent environmental inspections, one equipment from each successive 100, or one per month if less than one hundred are produced monthly, shall be selected. Equipments shall not be intermixed for sampling purposes, but shall be based on successive quantities of like equipments.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the quality assurance representative does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.4).

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to, and have passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.4.3.4 Temperature. Temperature tests shall be in accordance with the class 1 requirements of MIL-E-16400. The duty cycle under which the equipment shall operate during the temperature test shall be 5 minutes transmit to 10 minutes receive. During a one hour period at the end of 24 hours at -54°C and at the end of 4 hours at +65°C, 30 minutes at 90 percent input voltage and 30 minutes at 110 percent input voltage, the transmitter shall operate continuously. Reference measurements shall be made under standard conditions (see 3.5.1.1) in order to establish conformance with the following requirements throughout the temperature test (see 3.5.1.2).

4.4.3.4.1 Receiver.

4.4.3.4.1.1 Sensitivity shall not be degraded by more than 3 db (see 3.5.3.3).

4.4.3.4.1.2 Audio power output shall not be reduced by more than 4 db (see 3.5.3.12.3).

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4.4.3.4.2 Transmitter.

4.4.3.4.2.1 The carrier frequency shall not depart from the frequency measured under standard temperature conditions by more than:

+150°F ± 12 kHz

+100°F ± 10 kHz

+77°F ± 10 kHz

-40°F ± 15 kHz

-65°F ± 20 kHz

4.4.3.4.2.2 R.f. power output shall not be reduced by more than 3 db below the power measured under standard conditions at any temperature within the temperature range.

4.4.3.4.2.3 Noise modulation shall not be degraded by more than 6 db from the level under standard conditions at any temperature within the temperature range.

4.4.3.4.2.4 The channel changing cycle shall not exceed 20 seconds at any temperature within the specified range.

4.4.3.4.2.5 The transmit receive interval shall not exceed 406 ms. at any temperature within the specified range. Receive-transmit interval shall not exceed 70 ms. for the AN/SRC-21A and 130 ms. for the AN/SRC-20A.

4.4.3.4.2.6 Broadband sidetone amplifier. Measurement shall be made to determine conformance to 3.4.6.3.

4.4.3.4.2.7 Frequency modulation. Measurements shall be made to determine conformance to frequency modulation characteristics.

4.4.3.4.3 Radio Set Control C-3866/SRC. Radio Set control shall operate Radio Set AN/URC-9A in conjunction with suitable external chamber instrumentation that provides remote operation of the radio set.

4.4.3.5 Humidity. Humidity tests shall be in accordance with MIL-E-16400. Reference measurements shall be made under standard conditions (see 3.5.1.1) in order to determine conformance with the following requirements during the humidity test (see 3.5.1.2).

4.4.3.5.1 Receiver.

4.4.3.5.1.1 Sensitivity shall not be degraded by more than 3 db (see 3.5.3.3).

4.4.3.5.1.2 Audio power output shall not be reduced by more than 4 db (see 3.4.3.12.3).

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4.4.3.5.2 Transmitter.

4.4.3.5.2 R.f. power output shall not be reduced by more than 3 db.

4.4.3.5.2.2 Noise modulation shall not be degraded by more than 6 db.

4.4.3.5.3 Radio Set Control C-3866/SRC. Radio set control shall operate radio set AN/URC-9A in conjunction with suitable external chamber instrumentation that provides remote operation of the radio set.

4.4.3.6 Electromagnetic Compatibility test. Test shall be performed to determine conformance to 3.1.4. As a part of quality conformance inspection, this test shall be performed on the first sample equipments for environmental inspection and need not be performed again unless changes occur which might affect the measured characteristics.

4.4.4 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of section 5 of this specification.

4.5 Reliability testing. The reliability of the AN/SRC-20A, AN/SRC-21A, and AN/URC-9A shall be verified and reported in accordance with MIL-STD-781. The minimum acceptable MTBF for each equipment shall be 500 hours.

4.5.1 Qualification phase of production reliability tests. The qualification phase of reliability testing shall be conducted on the initial production units. Testing shall be performed at test level B for test plan IV. Input voltage cycling is required. Vibration acceleration shall be ±1 gravity. The sample size shall be in accordance with MIL-STD-781. The reliability requirements must be successfully demonstrated prior to the shipment of an production equipment.

4.5.2 Sampling phase of production reliability tests. The sampling phase of reliability testing shall be conducted throughout production to ensure continued production of acceptable equipment. Sampling testing shall be performed at test level B with test plan IV. Input voltage cycling is required. Vibration acceleration shall be ±1 gravity. The reliability test must be successfully completed prior to the shipment of production equipment. The sample size shall be in accordance with MIL-STD-781.

4.5.3 Reliability test plan. The contractor shall prepare a detailed reliability test plan in accordance with MIL-STD-781 (see 6.2). Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in the equipment specification. Performance parameters to be measured during the reliability test shall be in accordance with the equipment specifications and shall be measured at least twice each 24 hour period. Debugging and burn-in prior to reliability testing shall be limited to the debugging and burn-in to be performed on all production equipment to be delivered to the Government. No preventative maintenance shall be performed during the reliability test.

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of function must actually include all the defects listed under

4.5.4 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification. All failures are relevant failure unless determined by the procuring activity to be caused by a condition external to the equipment under test. Relevant failures include but are not limited to the following:

(a) A condition in the blower assembly whereby any required operating capability of the equipment changes beyond specified limits.
(b) A condition caused by equipment manufacturer workmanship or component failure whereby any required operating capability of the equipment changes beyond specific limits. Only relevant failure shall be counted during the reliability test for an accept or reject decision.

4.5.5 Reliability test reports. The contractor shall submit test reports on the procedures, results, test data and conclusions for the reliability qualification test and the reliability sampling tests. All test data and failure reports shall be submitted with the reliability test reports (see 6.2).

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with section 5 and documents referenced therein. First article packs which contain level A packages shall be subjected to rough handling tests (5.1.1).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Drying shall be in accordance with MIL-P-116.

5.1.1.3 Preservative application. None.

5.1.1.4 Unit packaging. Each radio set shall be individually packaged in accordance with MIL-P-116, ensuring compliance with the General paragraph under Methods of Preservation (Unit Protection) and the Physical Protection paragraph therein, and meet the rough handling tests of MIL-P-116. Cushioning and dunnage shall be used to prevent physical or mechanical damage.

5.1.1.4.1 Major components. The receiver-transmitter, power supply, control, and amplifier shall be packaged in accordance with method IIa of MIL-P-11 . Desiccant shall conform to MIL-D-3464, type II or III.

5.1.1.4.2 Other items. Items not securely affixed in place, such as electron tubes, shall be removed and packaged individually in transparent packages in accordance with MIL-E-17555. Hardware shall be packaged separately from other items in accordance with method III of MIL-P-116.

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5.1.1.4.3 Equipment repair parts. Equipment repair parts or running spares shall be individually packaged in sealed transparent packages in accordance with MIL-E-17555. These items shall be packaged separately from other items within the pack in containers conforming to PPP-B-636 and identified to the applicable equipment. Repair parts lists shall be provided in accordance with MIL-E-17555.

5.1.1.4.4 Technical literature. Technical literature shall be packaged in waterproof heat-sealed, transparent bags and shall not be placed within the barrier material used to package any other items.

5.1.2 Level C. Each complete radio set shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Sets packaged as specified in 5.1 shall be packed in a prone position in containers conforming to PPP-B-601, Type overseas, Style optional (see 5.1.1.4.1). Packing containers shall have closed and sealed case liners confroming to MIL-L-10547, and shall be strapped in accordance with the appendix to PPP-B-601. Containers exceeding 250 pounds gross weight shall be modified to include a skid-type base.

5.2.2 Level B. Sets packaged as specified in 5.1 shall be packed as described in 5.2.1 except that alternatively the packing containers may conform to PPP-B-601, Type domestic.

5.2.3 Level C. Items packaged as specified in 5.1 shall be packed in containers acceptable to the common carrier and which will ensure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

5.3 Unitized Loads. When applicable, items packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147, whenever quantitites to one destination equal 40 cubic feet or more. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number.

5.4 Marking. In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129 (see 6.1). Marking shall include special marking for method II packs.

5.5 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified, loose-fill polystyrene shall not be used for packaging or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Levels of preservation and packaging, packing, and marking required (see section 5).

6.2 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to, the following:

(a) GFE (3.1.1.2, 3.1.4 and 3.1.7)
(b) Reliability program plan (see 3.7.2).
(c) Reliability prediction (see 3.7.3).
(d) Reliability reports (see 3.7.5).
(e) Quality plan (see 4.1.2).
(f) Test procedures (see 4.1.4).
(g) First article inspection report (see 4.3.1).
(h) Reliability test plan (see 4.5.3).
(i) Reliability test reports (see 4.5.5).

6.3 Waiver of first article requirements. Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3.2).

6.4.1 Initial shipment shall be contingent upon successful completion of environmental inspection.

6.5 Changes from previous issue. The extent of changes (deletions, additions, etc.) preclude the annotation of the individual changes from the previous issue of this document.

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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